



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI RAM AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

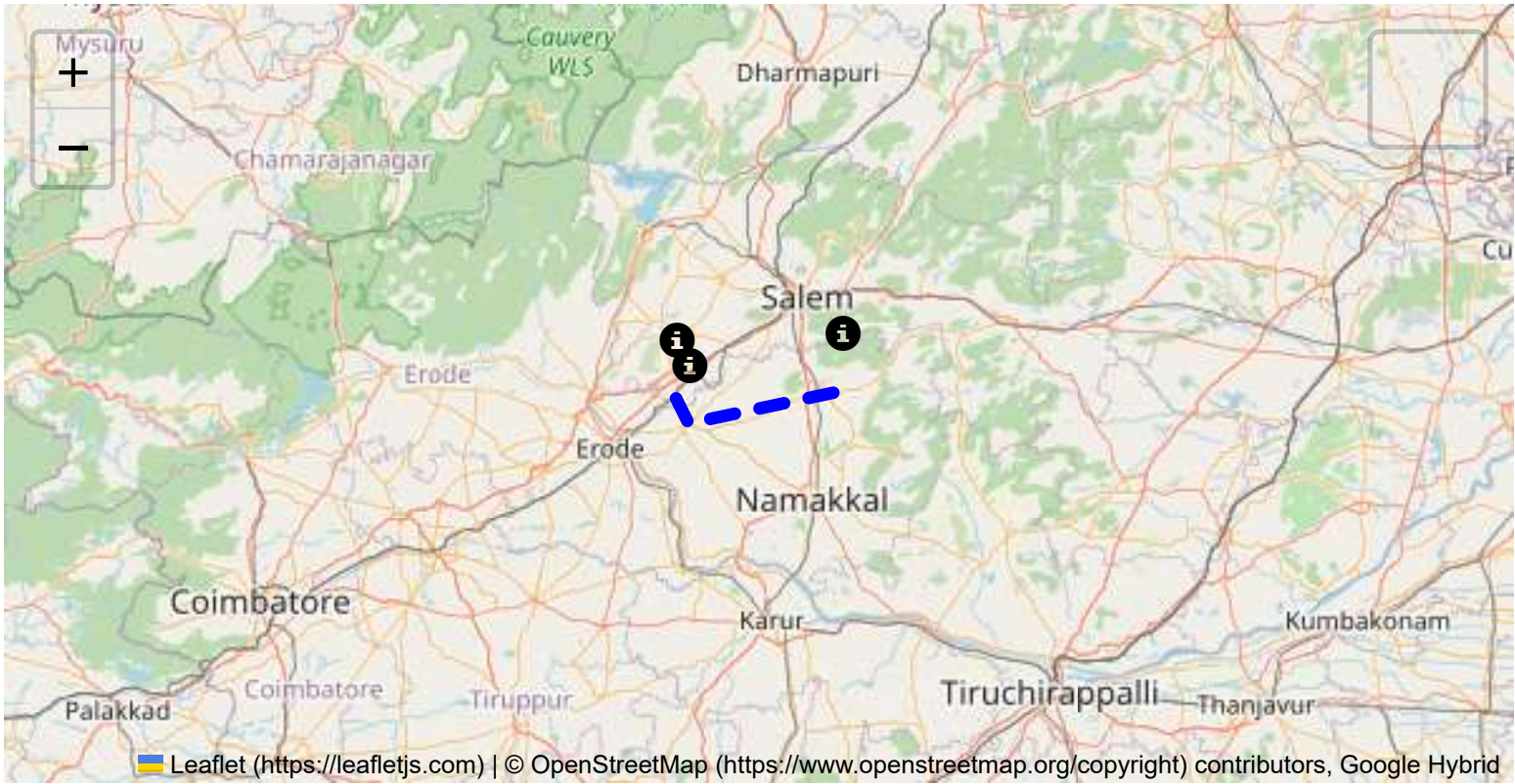
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

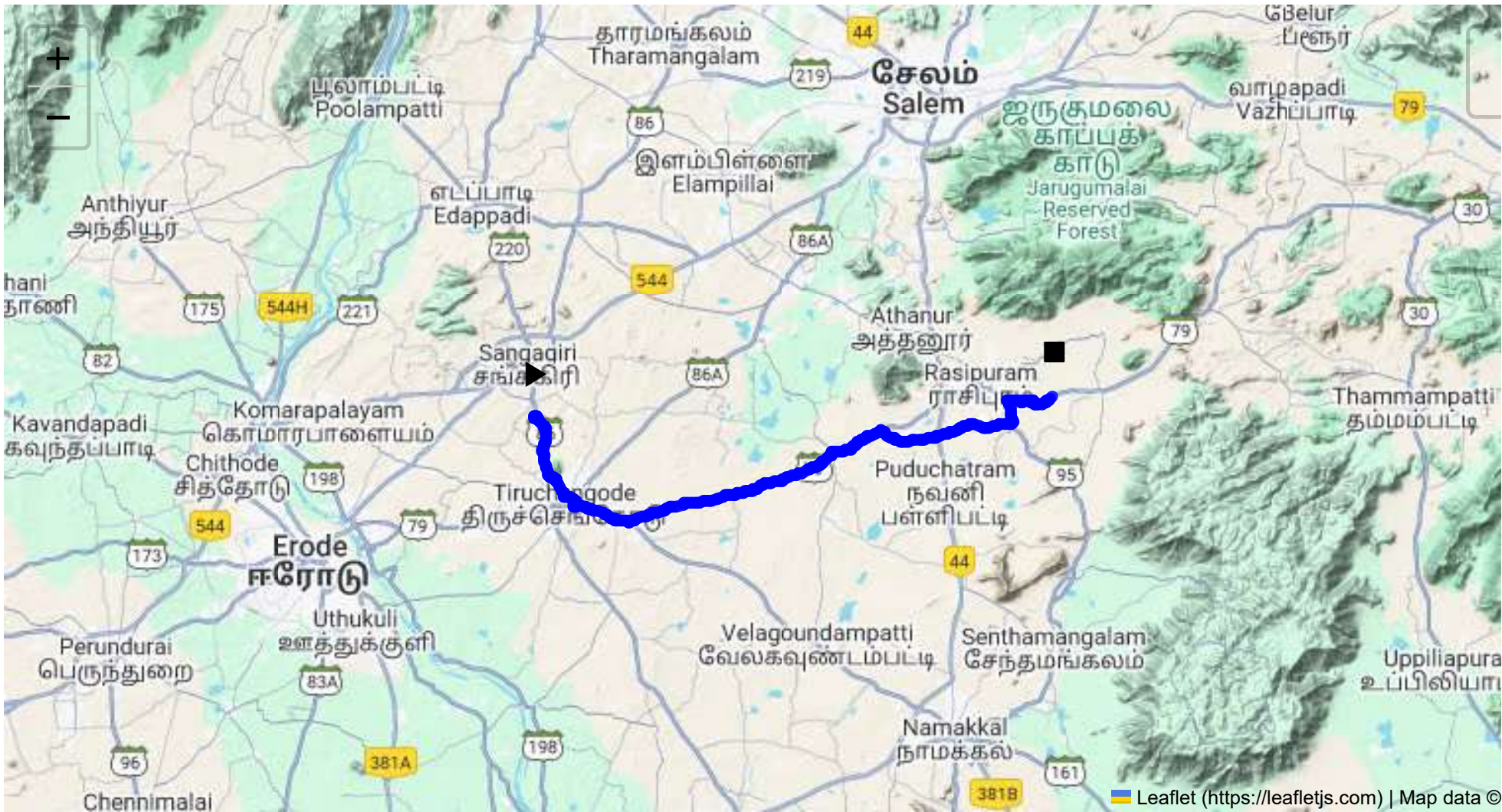
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 50.27 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.4525, 78.23482)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Vengayapalayam via Tiruchengode - Salem Main Road spans approximately 50.27 kilometers. The journey typically takes around 1.15 hours for heavy vehicles. This route traverses a mix of urban and rural areas, featuring both busy highways and narrower local roads.

Weather Conditions

Tamil Nadu experiences a tropical climate with monsoon seasons from June to September and October to December. During these periods, heavy rains can lead to slippery road surfaces and potential flooding, especially in low-lying areas. Heat in the summer months may also affect vehicle performance and road conditions.

Traffic Patterns

Traffic congestion is most common in the morning (7 AM to 10 AM) and evening (5 PM to 8 PM) during weekdays, especially near Tiruchengode. Weekends might also see increased traffic due to local markets and events. The Tiruchengode - Salem Main Road is a significant thoroughfare and might experience heavy vehicle traffic, impacting travel times.

Road Quality and Infrastructure

The major roads along this route are generally well-maintained, but some segments, particularly those passing through smaller towns or rural areas, might be narrower with less frequent maintenance. Potholes and uneven surfaces could pose risks during the monsoon season.

Alternative Routes

In case of emergencies, an alternative route via the NH44 towards Salem and then rejoining the local roads to Vengayapalayam can be considered, adding some distance but potentially offering better road conditions and increased access to facilities.

Local Regulations on Hazardous Materials

Transporting hazardous materials in Tamil Nadu requires adherence to specific guidelines, such as obtaining necessary permits and ensuring proper vehicle labeling. Drivers should be aware of designated routes for hazardous materials, especially in urban areas.

Historical Incidents

There have been occasional incidents involving heavy vehicles, including overturned trucks and minor collisions on this route. These have often been attributed to speeding, road conditions, and sudden braking.

Environmental Considerations

Certain stretches are near agricultural lands and communities, necessitating caution to prevent any contamination or spillage of hazardous materials. Awareness of sensitive zones, like schools or religious sites, is crucial.

Communication Coverage

Most of the route enjoys good mobile network coverage, but intermittent dead zones might occur, particularly in more rural segments. Ensuring reliable communication devices are available is advisable.

Emergency Response Times

In urban areas near Tiruchengode, response times are typically around 15-30 minutes, improving closer to major towns. In rural portions of the route, response times could extend to 45 minutes or more due to distance and road conditions.

Overall Risk Assessment

This route involves moderate risk for heavy trucks carrying hazardous materials. Risks stem from weather conditions, mixed traffic, and varying road infrastructure quality. Careful planning, adherence to regulations, and preparedness for potential communication and emergency delays are essential for safe transit.

In summary, thorough preparation and adherence to safety protocols will mitigate most risks on this route. Regular updates from local traffic authorities and weather reports can further enhance safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	Medium	11.38346, 77.89975	30 KM/Hr
5	Blind Spot	Blind Spot	11.38188, 77.90257	10 KM/Hr
6	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
7	Turn	Medium	11.41332, 78.08072	30 KM/Hr
8	Turn	Medium	11.41463, 78.08067	30 KM/Hr
9	Turn	High	11.42964, 78.11587	15 KM/Hr
10	Turn	High	11.43542, 78.20754	15 KM/Hr
11	Blind Spot	Blind Spot	11.45100, 78.20385	10 KM/Hr
12	Turn	Medium	11.44925, 78.22288	30 KM/Hr
13	Turn	Medium	11.44813, 78.22305	30 KM/Hr
14	Turn	Medium	11.44793, 78.22320	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.38188, 77.90257



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41332, 78.08072



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43542, 78.20754



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45100, 78.20385



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44925, 78.22288



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44813, 78.22305



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44793, 78.22320

